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Limitations in Reservoir Characterization in Stacked Carbonate Reservoirs, Supported by Seismic Inversion and Modelling in a Mature Oil Field

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Abstract

Unlike clastic rocks, stacked carbonate rocks present more challenges in reservoir characterization. This study highlights the results of seismic inversion, and the limitations encountered in reservoir characterization, focusing on the main producing carbonate reservoir in a mature offshore oil field.

Reservoirs in the study area are shallow water platform carbonates, covering part of inner to outer ramp deposition. Broadly, it is a coarsening upward succession of variable thicknesses of stacked reservoirs with different oil water contacts (OWC's). Two case studies are presented: the first uses a Pre Stack Time Migration (PSTM) seismic volume for porosity prediction through post-stack seismic inversion, while the second employs simultaneous inversion of angle stacks to determine P-wave impedance which was converted into porosity based on rock physics analysis. Both studies partially correlate seismic derived porosities with well based porosities. The study discusses five limiting factors causing discrepancies, including rock properties variations, tuning phenomena, the impact of fluid changes, overburden on seismic response and methodology. The study encompasses wedge modelling, forward modelling, Amplitude variation with Offset (AVO) response, fluid substitution modelling and seismic attribute maps to assist in identifying these limitations.

One of the primary challenges in reservoir characterization is managing thin reservoirs that are 5 to 10 feet thick and beyond seismic resolution. For thicker carbonate reservoirs, ranging from 60 to 120 feet, various other features influence the outcomes of reservoir characterization, such as high permeability streaks, relatively tight zones, and regions that are vuggy, fractured, or heterogeneous. Two seismic inversion studies presented show general agreement for porosity prediction in some reservoirs however few discrepancies were noted. Five limiting factors have been identified which individually or as combination cause discrepancies. Seismic response at reservoir is mainly affected by the overburden noise. The wedge model presented effectively determines vertical resolution (86ft with 25% porosity), which appears to change spatially due to porosity variation, indicating that not all reservoir units are seismically resolved. Forward models for several wells demonstrate a strong correlation with surface seismic data and capture changes in response with reservoir thickness and facies, which could significantly affect inversion results. Fluid substitution results confirm that displacing approximately 90% of oil with sea/saline water or gas results in at least 10% change in impedance and therefore on the derived porosities. For some reservoirs,

seismic attribute maps show poor correlation with well based porosity map, mainly because of combined effect of reservoir thickness, property and fluid change across the field.

The integrated approach presented in this paper emphasizes the seismic inversion study and highlights potential limitations in carbonate reservoir characterization, supported by various models and real datasets. Additionally, this paper explains how modelling results can aid in accurately predicting reservoir properties, ultimately leading to successful well optimization.

Introduction

The reservoirs in the study area are shallow water platform carbonates, spanning from inner to outer ramp deposition. Generally, it is a coarsening upward succession of stacked reservoirs with varying thicknesses and different oil-water contacts (OWCs). The study area is located offshore in the Middle East (Fig. 1). Structurally, most of the oil-producing fields are low dip (~ 1 -2 degrees) anticlines. The overburden section consists of a mix of anhydrite, calcite, and dolomite, with rocks mainly being saline aquifers. The reservoir rocks are primarily limestones, separated by dolomite layers, and some of the reservoirs are fractured. Present-day subsurface structures are mainly formed due to two tectonic episodes: one in the Late Cretaceous and another in the Late Oligocene. The main faults occurred in the Late Cretaceous age, trending primarily in the NW-SE direction and extending several kilometers. However, the tectonic stresses turned orthogonal (NE-SW) in the Late Oligocene age, further deforming the subsurface structure. As a result, subsurface complexity increased significantly due to changes in stress orientation over time (Ahmad et al, 2022).

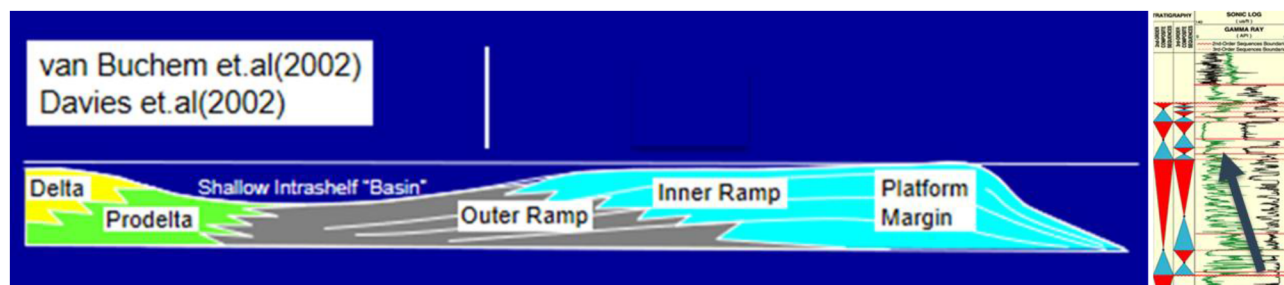


Figure 1—Reservoirs in the study area are inner to outer ramp deposition. Generally, it is coarsening upward succession of stacked carbonate reservoirs with varying thicknesses, as evident on the logs

The seismic response varies considerably due to many factors, including thin reservoir effects, thickness changes, facies/reservoir rock type (RRT) changes, depth of burial, porosity, fluid content, and velocity. It is important to consider these factors to establish a correlation between seismic information and rock properties. This paper presents two case studies focusing on porosity distribution in the reservoirs. The study also lists five limiting factors that potentially caused a mismatch between estimated porosity and actual porosity from the logs and core data.

Seismic data overview

In 2001, wide azimuth 2C (hydrophone and geophone) OBC (Ocean Bottom Cable) data covering the entire field was acquired, with a bin dimension of $12.5\text{m} \times 25\text{m}$. The data was processed in 2002 (Old) but the data quality was not good. To obtain high-quality and high-resolution data, another reprocessing effort was completed in early 2009 (New). The objective was to produce seismic data suitable for structural and stratigraphic interpretation, quantitative reservoir characterization, and high-quality imaging of the concession's deeper target reservoirs. Compared to the previous processing, the new reprocessed PSTM volume represents a significant enhancement in imaging for structural interpretation. The noise attenuation

was significantly improved (except for some overburden noise, details is provided in the following section), achieving a much higher signal-to-noise ratio. Multiple attenuation was also improved, with very limited occurrence of residual multiples. A comparison of the processing results is shown in Fig. 2.

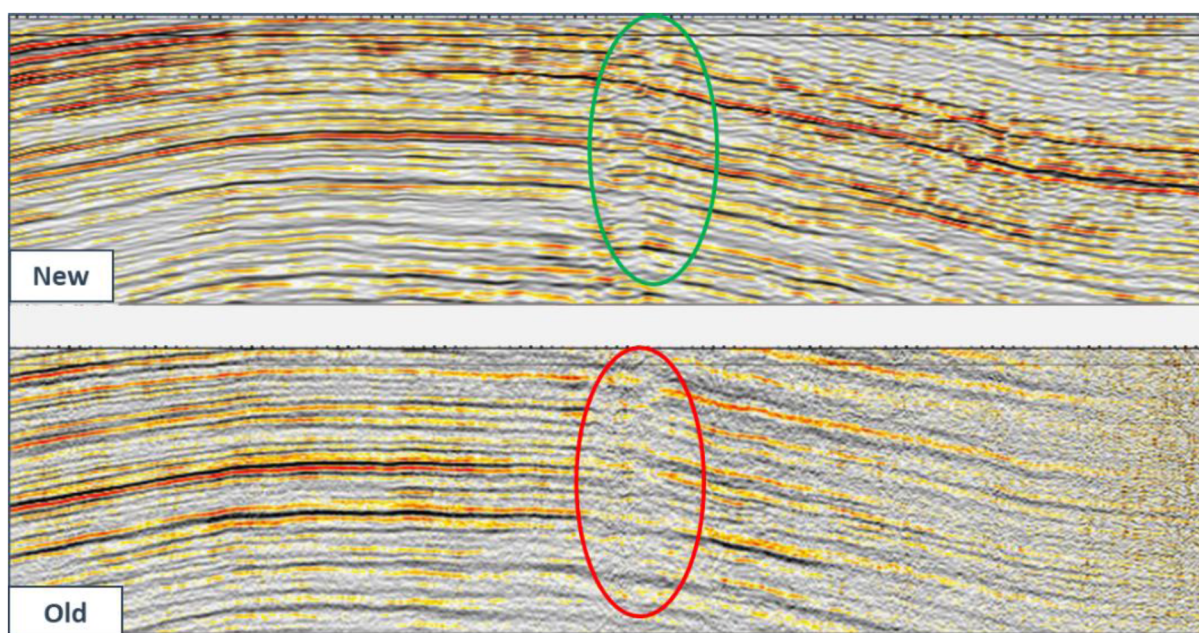


Figure 2—Comparison of old and new seismic processing. New seismic data show improvement in structure and enhancement in signals to noise ratio

Due to the geological complexity, significant lateral velocity variations were observed in the overburden section, leading to anomalies that extend down to the reservoir. This impacts not only the structure but also the attributes, such as amplitude, etc. Additionally, it was noted that the carbonate reservoirs in the study area exhibit various features that could influence the outcomes of reservoir characterization, including high permeability streaks, relatively tight zones, and regions that are vuggy, fractured, or heterogeneous. Besides changes in reservoir properties, lateral thickness changes were also observed in the reservoirs, resulting in tuning thickness variations laterally within a reservoir from crest to flanks. Due to geological variations and some remnant noise in the seismic data, attributes generated from the final stack show 70-80% conformance with reservoir properties (porosity) only in areas where the data is relatively cleaner.

Case Study#1

The new reprocessed Pre Stack Time Migration (PSTM) volume was used to predict porosity distribution in the reservoirs through post-stack seismic inversion. Constrained Sparse Spike Inversion (CSSI) was run on the full stack seismic to produce an impedance volume that was later converted into a porosity volume, based on the good correlation between the two attributes observed in wireline log data (Fig. 3). The results of the study indicate:

- Seismic response is honored (seismic constrained).
- Due to the limitations of seismic bandwidth, the focus was placed only on a few reservoir layers.
- Rock Physics analysis indicates a good correlation between porosity and V_p , impedance, etc.
- There is a good agreement between seismic-derived porosity and well-based porosity maps in most areas. However, a poor match was observed in a few local areas due to tuning and residual overburden effects.

- It was recommended to evaluate neural networks as another option for optimizing porosity prediction.

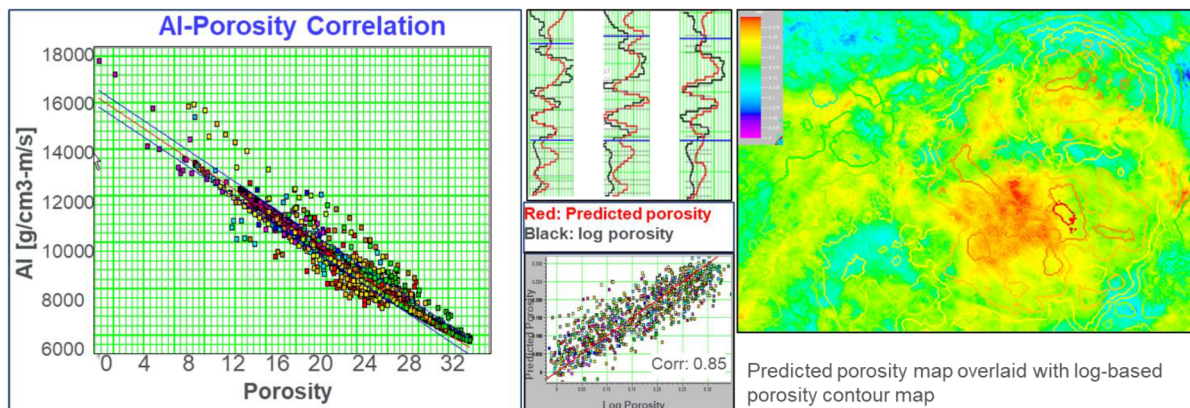


Figure 3—Porosity prediction and validation utilizing post-stack Constrained Sparse Spike Inversion (CSSI). The porosity map derived from seismic data, shown on the right, is overlaid with colored contours representing actual porosity measurements from logs and core samples. Discrepancies were noted in multiple locations.

Case Study#2

After obtaining meaningful outcomes from post stack deterministic acoustic impedance inversion results, a new study was conducted to evaluate, if pre-stack (i.e., angle stack) inversion can provide uplift to porosity and fluid prediction over post-stack inversion. The motivation factor was that the Pre-stack inversion produces smaller uncertainty in porosity prediction than post-stack inversion because it separates fluid effect from porosity effect in seismic data.

The simultaneous inversion inverts angle stacks to determine P-wave impedance, S-wave impedance and density. It is based on the assumptions that PP reflectivity as a function of angle can be modelled with the Aki-Richards equations (Aki and Richards, 2002). This study was conducted in a pilot area. The workflow applied in this study is a standard simultaneous inversion workflow. Multiple angle stack seismic data are integrated with well data through simultaneous inversion to produce a model of the absolute elastic properties of the reservoirs. Logs from 16 wells were analyzed to derive relationships between acoustic properties and rock properties, and several combinations of porosity and fluid saturation situation were modeled to derive pre-stack seismic responses in relation to these porosity and fluid saturation values. In addition, seismic data quality was evaluated to determine the applicability of porosity and fluid predictions.

Rock physics analysis indicated that pre-stack inversion might provide uplift in porosity prediction and possibly fluid discrimination when compared to post-stack inversion. Given relatively favorable rock physics results, a pilot pre-stack elastic inversion test was run using newly computed angle stacks with angle ranges which capture the amplitude variability from near to far angles. The resulting P impedance (I_p) cube was used as input into porosity predictions. Porosity map for the reservoir was analyzed and compared against the map from the validated geologic model (Fig. 4). The results of the study indicate:

- Honoring seismic response (seismic constrained)
- Pilot inversion shows general agreement with Impedance from well logs (up to ~75 Hz)
- Porosity prediction is not quite as rigorous as from previous studies using P Impedance which may be a result of noise in the inversion.
- Noise in seismic data partly an issue

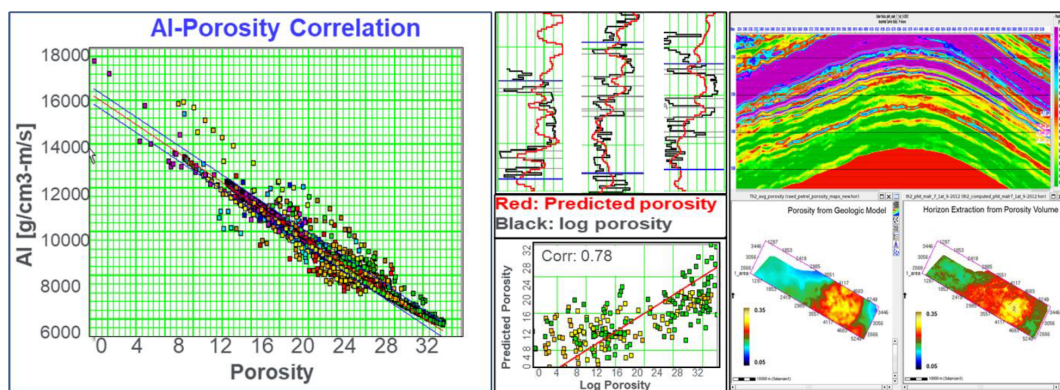


Figure 4—Porosity prediction and validation using Pre-stack inversion.
Predicted porosities show 78% correlation with log porosities.

Full field simultaneous elastic inversion was not recommended due to inconsistent results from the pilot study unless it is in conjunction with a high-end reprocessing effort in depth to further improve the imaging using the current dataset or until there is newly acquired seismic data over the Field.

Limiting Factor#1

Bulk density reflects the rock's compaction and porosity, making it useful for estimating porosity and fluid saturation. Bulk rock densities primarily vary due to porosity. Generally, higher bulk density indicates lower porosity and greater compaction (SEG WIKI). The study area encompasses a vast oil field with multiple stacked carbonate reservoirs. It has a vertical relief of approximately 1100 feet and extends laterally about 60 kilometers, with hundreds of well penetrations. Based on the analysis from over fifty well, a significant change in bulk density has been observed from the crest (low density) to the far flank (high density), with reservoir-2 ranging from 2.2 g/cm³ to 2.45 g/cm³ (Fig. 5). This indicates that reservoir porosity varies from the crest to the flanks, as evidenced by the actual porosity map of reservoir-2. The impact of porosity change is observed in various fields globally and can be significantly addressed if the reservoir property distribution is well understood. The heterogeneities are further amplified by the presence of natural fractures and dissolution features like vugs and high perm steaks. The variability in fluid and rock properties results in differing seismic responses, which can lead to challenges in establishing a reliable quantitative correlation between seismic data and reservoir characteristics in certain reservoirs and regions.

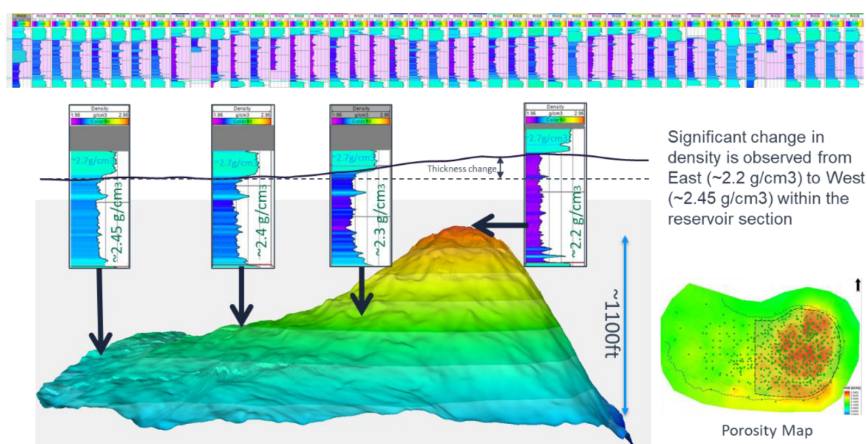


Figure 5—Reservoir property changes across the field, showing significant density variation from crest to flank. The log-derived porosity map presents the distribution of porosities for reservoir-2.

Limiting Factor#2

Tuning is a common phenomenon in seismic data associated with thin beds. It involves the brightening or dampening of seismic amplitude due to constructive and destructive interference from overlapping seismic reflectors (SEG Wiki). Tuning is typically illustrated using a wedge model that shows seismic response changes based on layer thickness. Fig. 6 presents a typical wedge model with a reservoir having average porosity of 25% (ranging from 20% on the flanks to 32% at the crest), interval velocity from well logs, and a wavelet with a maximum frequency of 50Hz. The distinct reflections and the thickness at which interference begins are marked by a black arrow, indicating one-quarter of the wavelength ($\lambda/4$), which corresponds to a thickness of 86 feet. The seismic cross-section example on the right (Fig. 6) shows that reservoir-2 is thick enough to generate a distinct seismic response without interference. Although reservoir-3 is 60 feet thick, it still produces a distinct seismic response (trough) at field's crest mainly due to its 32% porosity, which reduces the tuning thickness to approximately 60 feet. On the flanks, since the porosity decreases to around 20%, increasing the tuning thickness to over 100 feet. Consequently, the reservoir that generates a distinct seismic response at the crest will not produce a distinctive response at the flank due to the tuning thickness. Furthermore, as the reservoir thins towards the flank, the seismic response will be affected by the combination of changes in thickness and porosity. This complexity makes it more challenging to correlate the seismic response with properties derived from well logs.

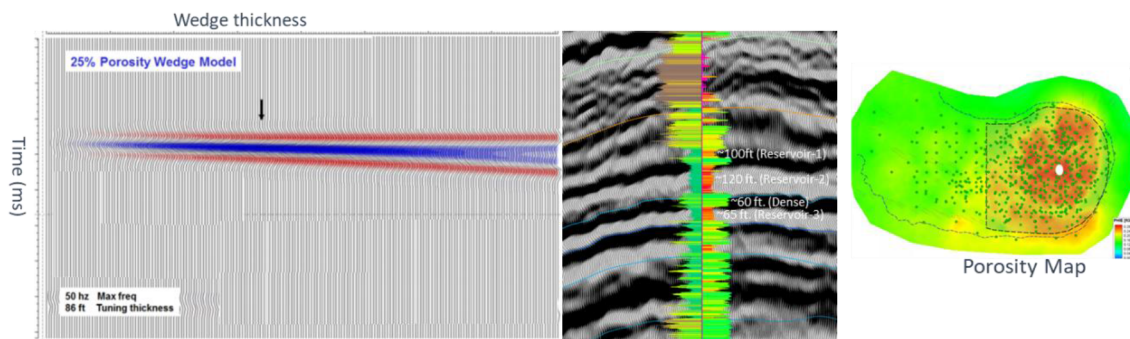


Figure 6—Wedge modeling represents the thickness at which interference begins is marked by a black arrow, indicating one-quarter of the wavelength ($\lambda/4$), which corresponds to a thickness of 86 feet.

When the reservoir-2 forward models are generated for several wells, they exhibit changes in seismic response, such as a clear doublet for the wells on the crest and a single trough for the wells on the flanks. These models show a strong correlation with the surface seismic data. A limiting factor in such reservoirs is identifying the effect of tuning thickness. Without this, any attributes extracted from the surface seismic may be misleading. As shown in Fig. 7, the amplitude extractions from the autotracked horizons highlight the lowest amplitudes on the flank (encircled in white), which do not conform with the actual porosity distribution shown in Fig. 6.

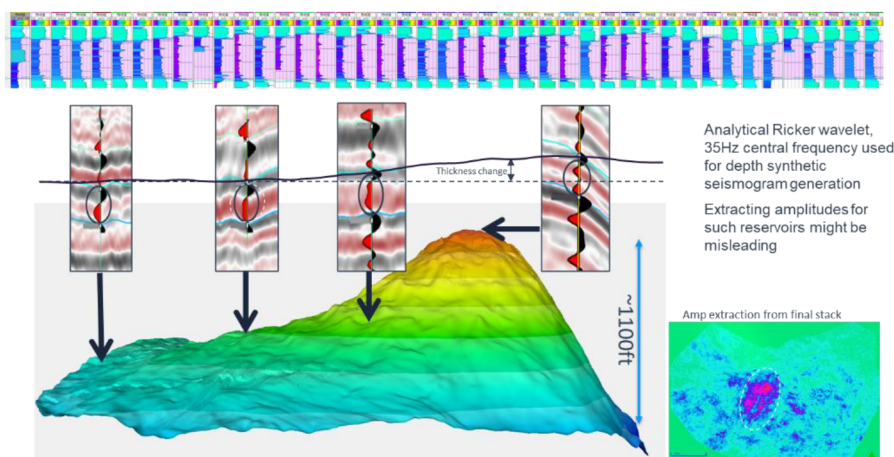


Figure 7—The forward models show a strong correlation with the surface seismic data. A limiting factor in such reservoirs is identifying the effect of tuning thickness. Without this, any attributes extracted from the surface seismic may be misleading.

Limiting Factor#3

In addition to changes in rock properties and thickness, the effect of fluid change has also been modeled, showing significant differences when the initial reservoir fluid is replaced with saline water or gas. Fluid substitution results confirm that displacing approximately 90% of oil with sea/saline water or gas results in an at least 10% change in impedance and therefore on the derived porosities. Since porosity is derived from impedance, the correlation coefficient between the two will be affected after fluid replacement. The accuracy in the correlation is also affected when porosity is derived from logs acquired 15 to 20 years ago, during which time many dynamic changes have occurred in the reservoir due to injection and production (Fig. 8).

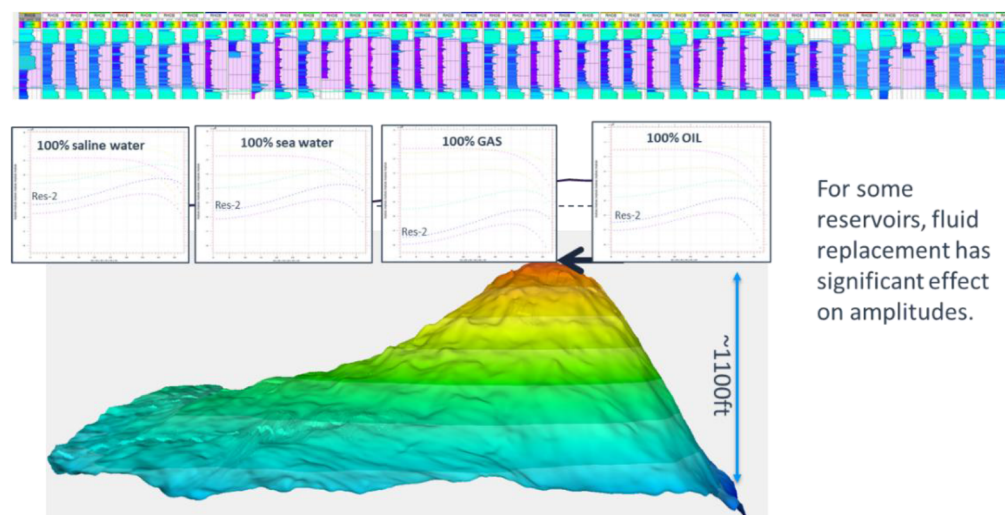


Figure 8—Fluid replacement effect on seismic amplitudes. Displacing approximately 90% of oil with sea/saline water or gas results in an at least 10% change in impedance and therefore on the derived porosity.

Limiting Factor#4

The impact of the overburden on the reservoir is more pronounced than the previously mentioned limiting factors. While the newly reprocessed PSTM volume offers a significant improvement in structural interpretation and better noise attenuation with a much higher signal-to-noise ratio, some noise persists in the data. This is primarily due to geological complexity; strong lateral velocity variations in the overburden section create anomalies that cascade down to the reservoir. These anomalies affect not only the structure but

also wavelet attributes, such as amplitude. Attributes from the affected amplitude zone, including derived porosities, will be misleading. Therefore, it is equally important to analyze overburden and associated effects on reservoir. Examples of these effects, including karst collapse features, channels, and strong footprints (not removed from the seismic during processing stage), are illustrated in Fig. 9.

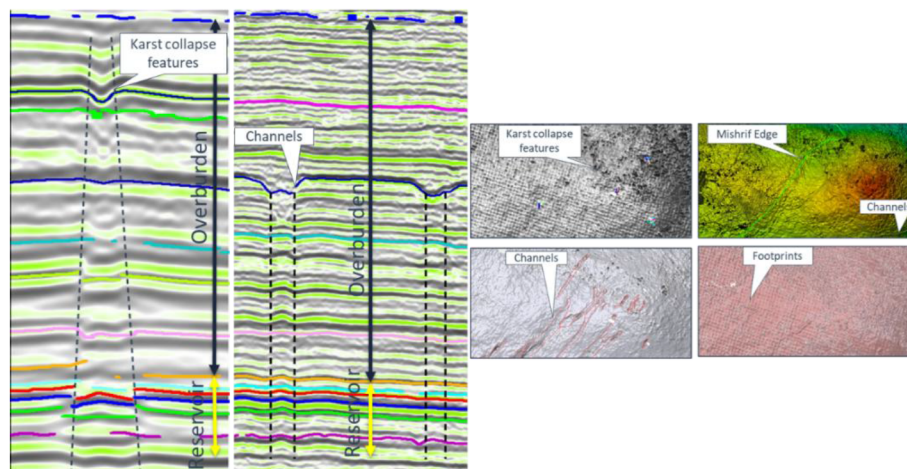


Figure 9—Effect of overburden on the reservoir effects structure and wavelet attributes, such as amplitudes. Attributes from the affected amplitude zone, including derived porosities, will be misleading.

Limiting Factor#5

The selection of a seismic inversion method is crucial as it affects the accuracy, resolution, and reliability of results in reservoir characterization. Various methods offer distinct advantages and limitations based on data quality, study objectives, and geological complexity (Saeed Izadian, 2024). For instance, a band-limited inversion technique may be appropriate for estimating acoustic impedance swiftly in relatively simple geological environments, although it might fail to capture finer scale variations. Conversely, geostatistical inversion can predict porosity distribution within a reservoir with greater precision and provide uncertainty estimates; however, it demands higher computational resources, particularly when handling large datasets.

Future Considerations

In the presence of strong lateral velocity variations in the overburden, an earth image derived from time migration is not accurate, making depth migration (PSDM) essential for obtaining an accurate earth image. Moreover FWI (Full Waveform Inversion) is expected to resolve overburden anomalies. Combining FWI-based PSDM with standard inversion methodology is expected to produce reasonable results that match actual porosity. Geostatistical inversion can be considered to predict porosity distribution with greater precision, especially for the thin reservoirs.

Conclusion

The study addresses the challenges in characterizing stacked carbonate reservoirs, focusing on seismic response and the correlation between seismic information and rock properties. Two case studies are presented: the first one uses a reprocessed final Pre Stack Time Migration (PSTM) volume for porosity prediction through post-stack seismic inversion, and the second employs simultaneous inversion of angle stacks to determine P-wave impedance which is converted into porosity based on the result of rock physics. Both studies partially correlate estimated porosities with actual porosities, though discrepancies due to tuning, rock properties, fluid properties and overburden effects are noted. The study discusses five limiting factors, including rock properties variations, tuning phenomena, the impact of fluid changes, overburden

on seismic response and the choice of algorithm. Ultimately, the study emphasizes the need for depth migration (PSDM) to obtain accurate earth images in the presence of strong lateral velocity variations in the overburden, suggesting that combining FWI-based PSDM with standard inversion methodology can yield accurate porosity predictions.

Acknowledgments

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Nomenclature

AVO	=	Amplitudes variations with Offset
CSSI	=	Constrained sparse spike inversion
FWI	=	Full Waveform Inversion
IP	=	P Impedance
OBC	=	Ocean Bottom Cable
OWC	=	Oil Water Contact
PSDM	=	Pre Stack Depth Migration
PSTM	=	Pre Stack Time Migration
RRT	=	Reservoir Rock Type

References

- Ahmad, J., Khouri, N., Sarra, A., Khulood, A., 2022, Overburden drilling challenges & their possible mitigations using integrated seismic structure characterization in a giant carbonate field in Middle East, Abu Dhabi International Petroleum Conference and Exhibition, 31 October-03 November 2022, Abu Dhabi, UAE, SPE 211657.
- Aki, K., and Richards, P.G. (2002). *Quantitative Seismology*, 2nd Edition. W.H. Freeman and Company.
- Saeed Izadian, 2024, Tuning effect in time-lapse seismic inversion for CO₂ plume monitoring at Sleipner field, *International Journal of Greenhouse Gas Control* **137** (2024) 104224, UK.
- SEG Wiki, <https://wiki.seg.org/wiki/>